Requirements Analysis Document

Domain

Online Gaming and Interactive Entertainment

Problem Statement

Online Multiplayer Game Management Platform (OMGMP)

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1. Preface

1.1 Expected Readership

This Software Requirements Specification (SRS) document is intended for faculty evaluators, project reviewers, system analysts, designers, and developers involved in the academic development of the **Online Multiplayer Game Management Platform (OMGMP)**. The document is written in clear and simple English so that readers with varying technical backgrounds can easily understand the system scope, actors, and requirements.

1.2 Version History

- **Version 1.0** – Initial version of the Requirements Analysis Document.
  This version documents the finalized problem statement, actors, features, and system requirements identified during the requirements analysis phase. It establishes a baseline before moving to design-level artifacts such as UML diagrams.

1.3 Summary

This is the first version of the document; therefore, no changes from previous versions exist. Future revisions may include refinements based on faculty feedback, expanded system scope, or changes in project understanding.

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2. Introduction

2.1 Need for the System

Modern online multiplayer environments involve multiple players, games, sessions, and competitive activities operating simultaneously. When player participation, session handling, game availability, and rule enforcement are managed in a fragmented or unstructured manner, it becomes difficult to ensure fair gameplay, proper monitoring, and consistent user experience.

An **Online Multiplayer Game Management Platform** is required to provide a centralized system that manages player access, multiplayer sessions, game lifecycle control, and administrative oversight. The platform simplifies coordination between players and platform authorities while maintaining structured control over gameplay activities.

2.2 System Overview and Functions

The OMGMP acts as a centralized platform that supports multiple roles and manages multiplayer gaming activities across different games. The system provides functionality such as:

- Player authentication and profile management
- Game selection and participation
- Creation and management of multiplayer sessions (rooms, matches, tournaments)
- Session monitoring and rule enforcement
- Storage and access to past session and match history
- Administrative control over users, games, and platform policies

The system ensures that interactions between players, moderators, publishers, and administrators follow defined workflows.

2.3 Interaction with Other Systems

The platform may interact with supporting systems such as:

- Player account and profile data storage
- Game session servers (conceptual)
- Notification services for alerts and updates

These interactions support authentication, session tracking, and communication within the platform.

2.4 Business and Strategic Objectives

The Online Multiplayer Game Management Platform supports structured digital game governance by aiming to:

- Improve organization of multiplayer sessions and tournaments
- Ensure fair play through monitoring and enforcement
- Centralize game and user management
- Enhance transparency and accountability in online gameplay

The platform provides a scalable foundation for managing multiplayer gaming ecosystems.

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3. Glossary

Term	Definition
Online Multiplayer Game Management Platform (OMGMP)	A centralized system that manages online multiplayer games, sessions, and user roles
Player	An end user who participates in games and multiplayer sessions
Game Publisher	An entity responsible for listing, updating, and managing games on the platform
Moderator	A role responsible for monitoring sessions and enforcing platform rules
Platform Administrator	A role responsible for managing users, roles, and platform policies
Session	A single instance of multiplayer gameplay involving one or more players
Tournament	A structured competitive event involving multiple game sessions
Policy	A set of rules and guidelines governing platform usage and behavior
Enforcement Action	An action taken to handle rule violations, such as warnings or restrictions

4. User Requirements Definition

4.1 User Services (Functional Requirements)

4.1.1 Player

- The system shall allow players to register and authenticate.
- The system shall allow players to select and participate in available games.
- The system shall allow players to create or join multiplayer sessions.
- The system shall allow players to view match history and rankings.
- The system shall allow players to report other players for rule violations.

4.1.2 Game Publisher

- The system shall allow publishers to list and manage games on the platform.
- The system shall allow publishers to update game availability and metadata.
- The system shall allow publishers to monitor game usage statistics.

4.1.3 Moderator

- The system shall allow moderators to monitor active game sessions.
- The system shall allow moderators to handle player reports.
- The system shall allow moderators to apply enforcement actions.

4.1.4 Platform Administrator

- The system shall allow administrators to manage users and roles.
- The system shall allow administrators to manage platform policies.
- The system shall allow administrators to approve or suspend publishers and games.

4.2 Non-Functional User Requirements

- The system shall be easy to use and require minimal learning effort.
- The system shall provide acceptable response time during gameplay-related actions.
- The system shall ensure secure handling of user data.
- The system shall be available except during scheduled maintenance.

4.3 Product and Process Standards

- The system shall follow standard software engineering and UML modeling practices.
- The system shall comply with basic security and data handling principles.

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5. System Architecture

5.1 High-Level Architectural Overview

The platform follows a modular architecture consisting of:

- User Management Module
- Game Management Module
- Session and Tournament Management Module
- Moderation and Enforcement Module
- Notification and Reporting Module

5.2 Distribution of Functions Across Modules

- Players interact primarily with the Game and Session Management modules.
- Publishers interact with the Game Management module.
- Moderators interact with the Moderation module.
- Administrators interact with all modules through role-based access.

5.3 Reusable Architectural Components

- Authentication and Authorization Component
- Notification Component
- Reporting and Logging Component

This architecture supports scalability, clarity, and maintainability.

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6. System Requirements Specification

6.1 Functional Requirements

- The system shall support secure user authentication.
- The system shall manage multiplayer session lifecycles.
- The system shall store and retrieve session history.
- The system shall support monitoring and enforcement workflows.

6.2 Non-Functional Requirements

- **Performance:** The system shall handle multiple concurrent users.
- **Security:** The system shall protect user accounts and session data.
- **Reliability:** The system shall maintain consistent session state.
- **Usability:** The system shall be intuitive for all user roles.

6.3 External Interfaces

- Web-based user interface
- Notification interface (conceptual)
- Database interface for persistent storage

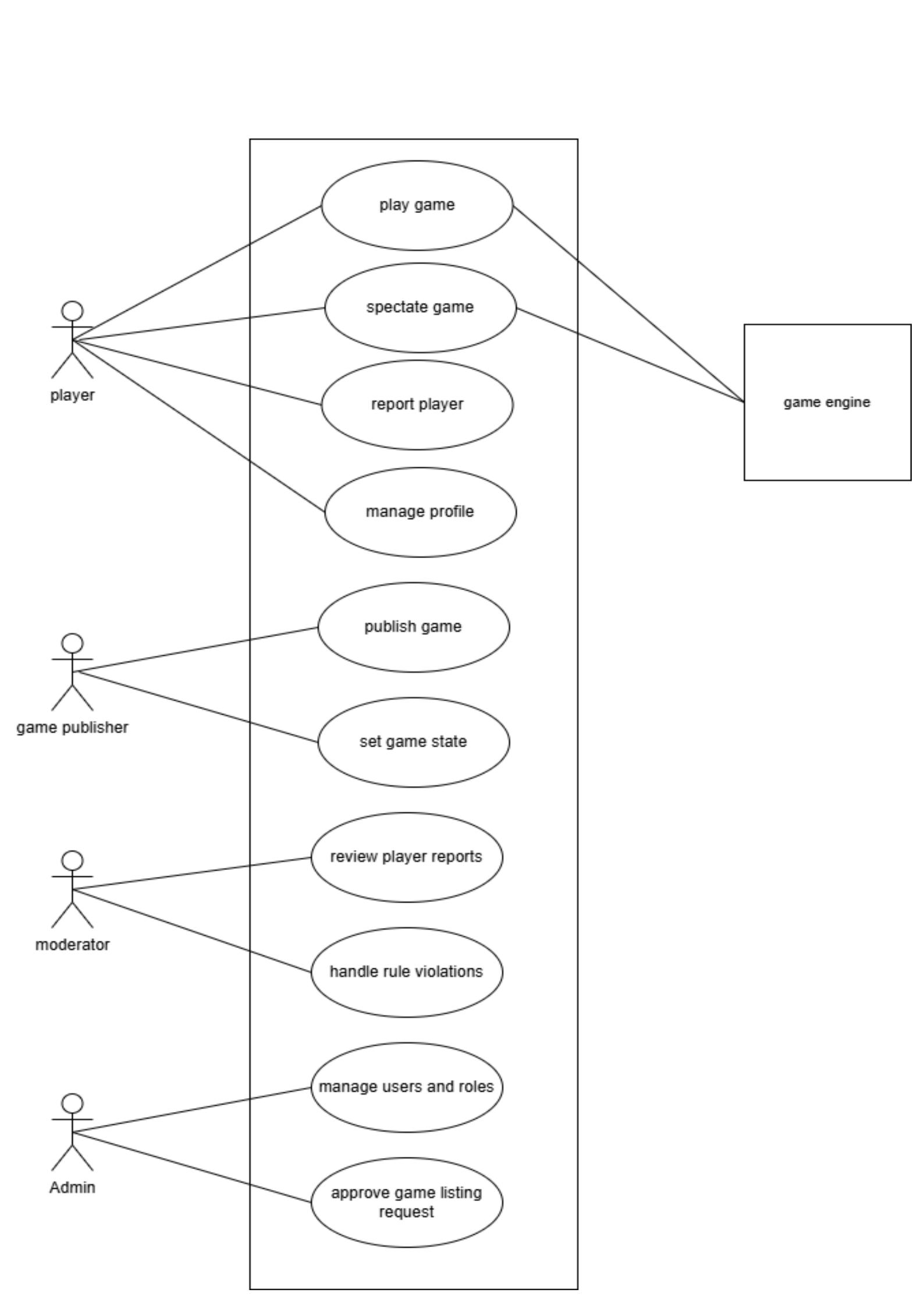
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7. System Models

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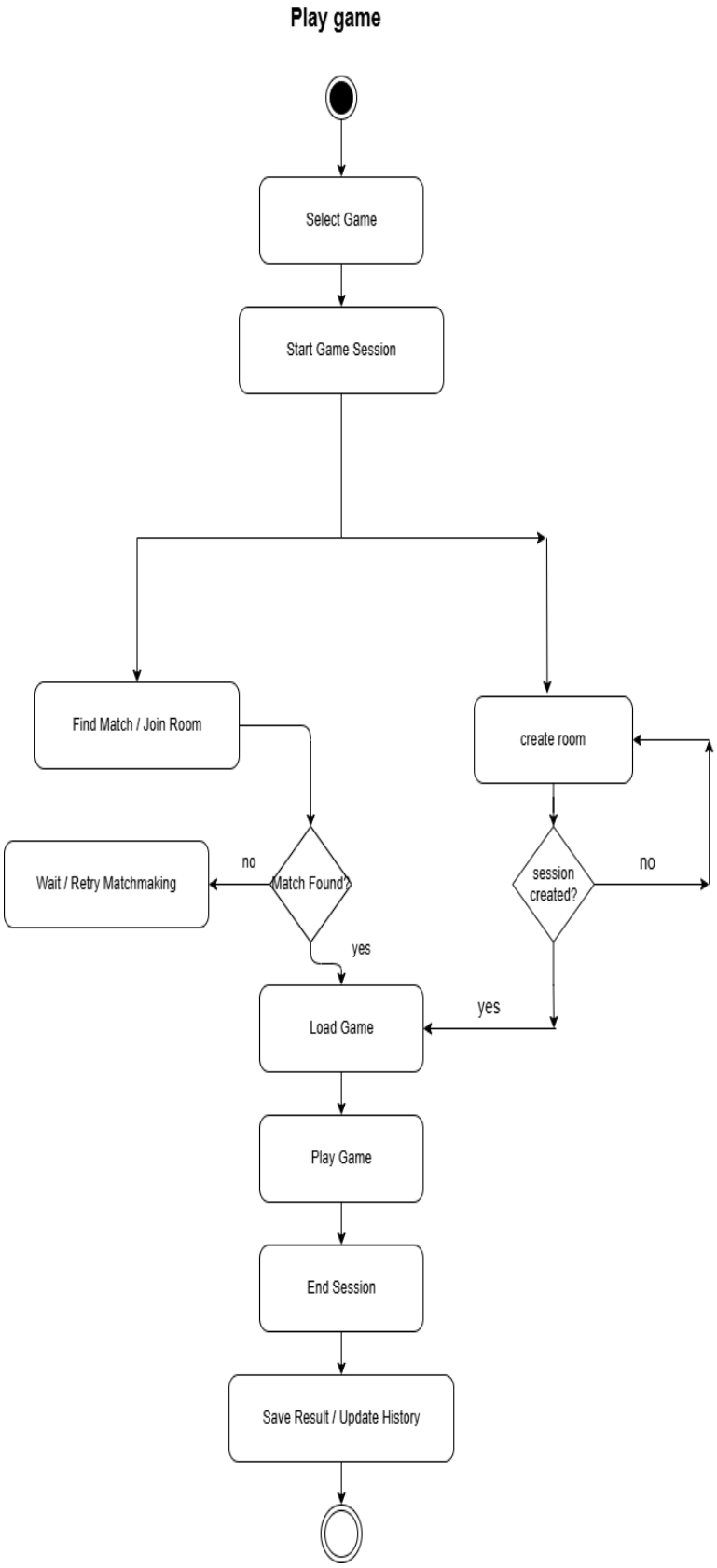
- Use case diagram
- Activity diagram
- Sequence diagram

Use case diagram

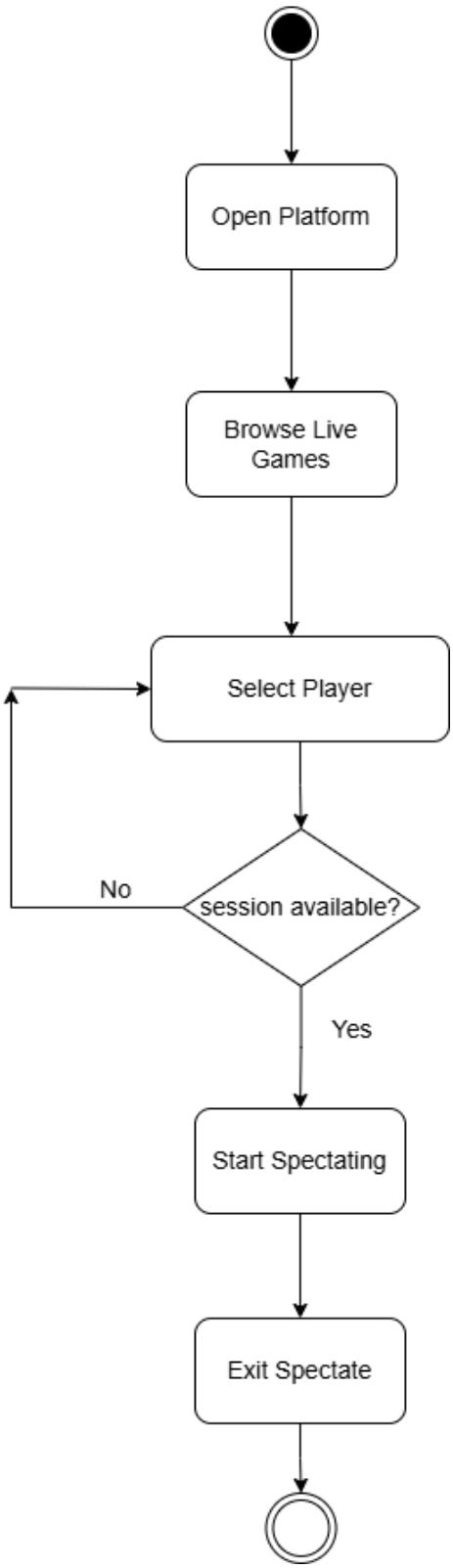

play game
spectate game
report player
manage profile
publish game
set game state
review player reports
handle rule violations
manage users and roles
approve game listing request
player
game publisher
moderator
Admin
game engine

Activity Diagrams

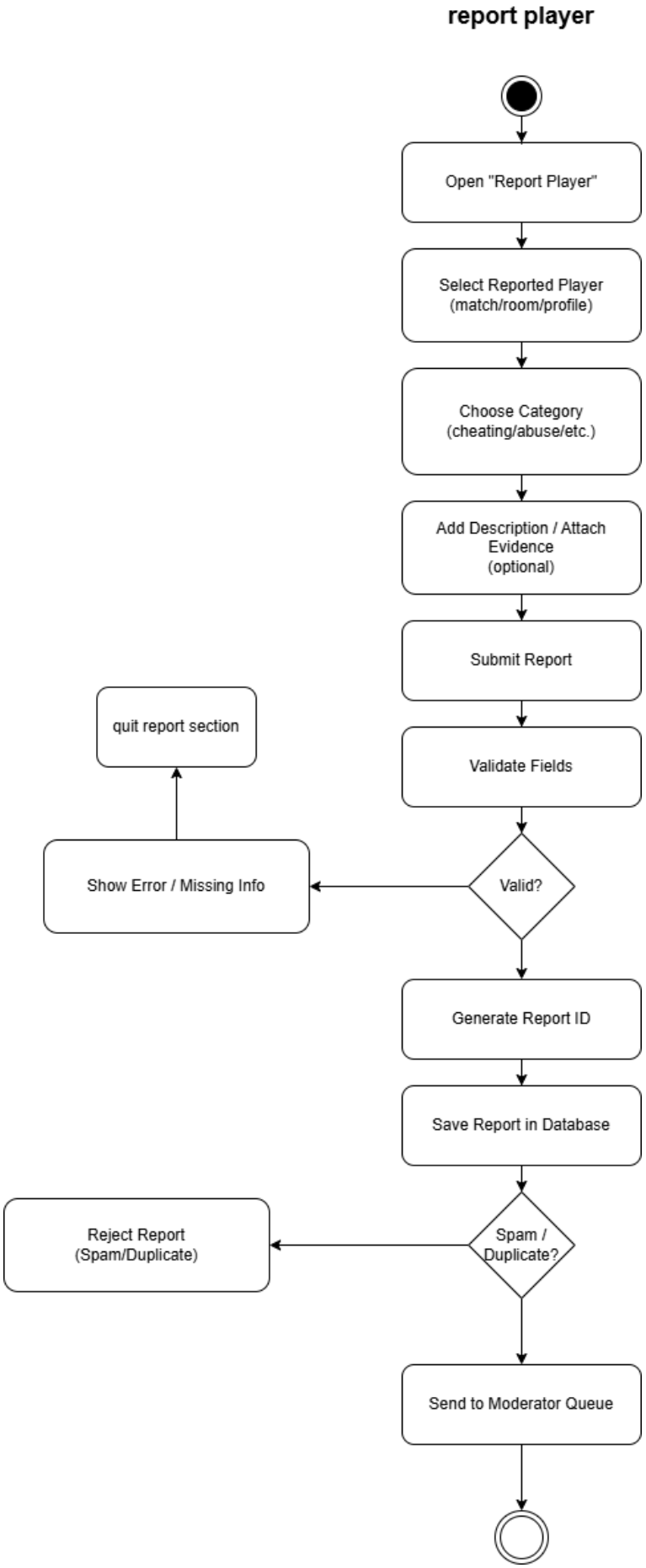
Activity Diagrams

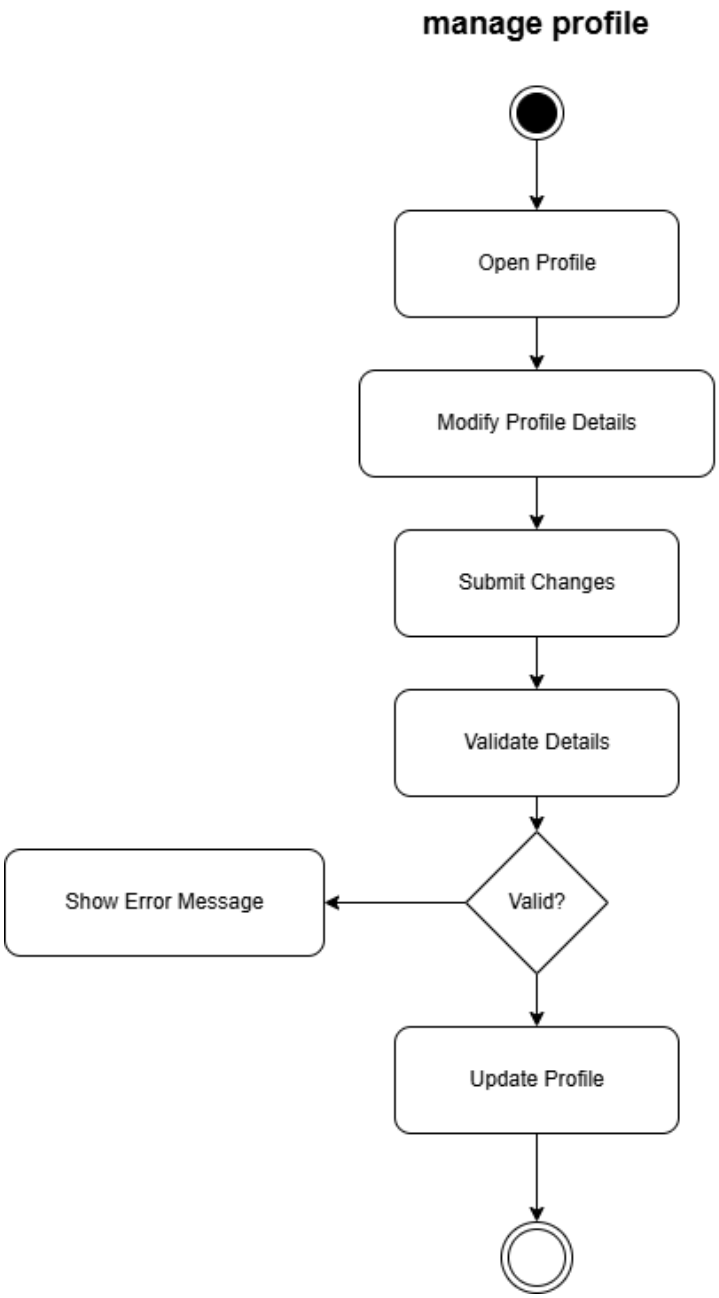


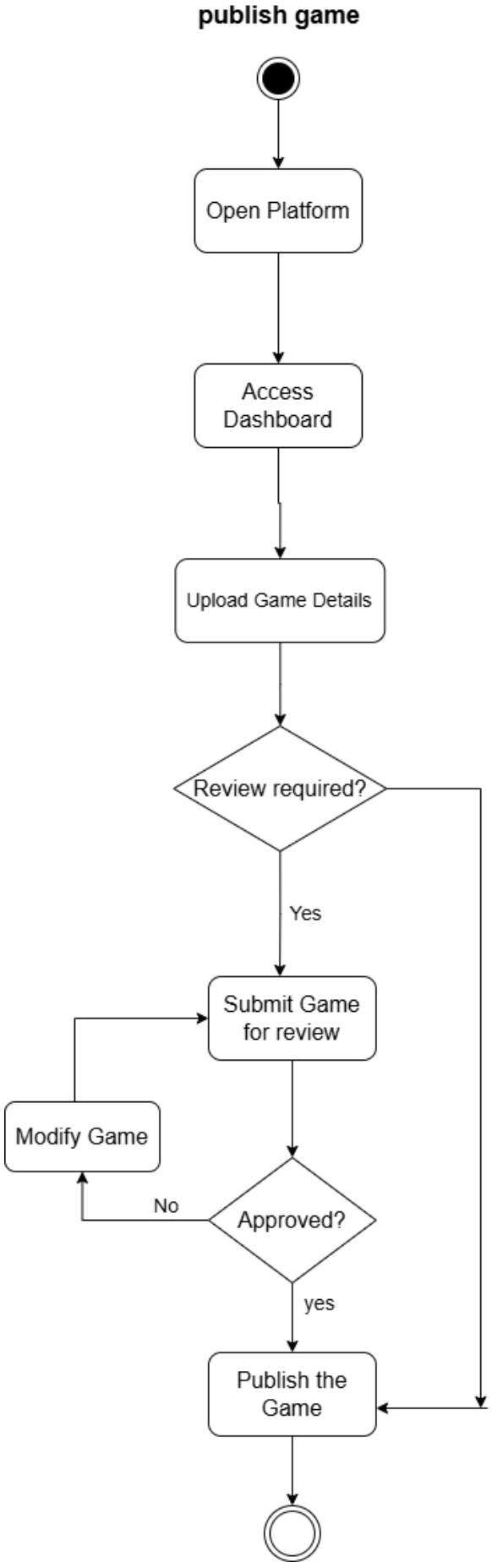
Spectate a game



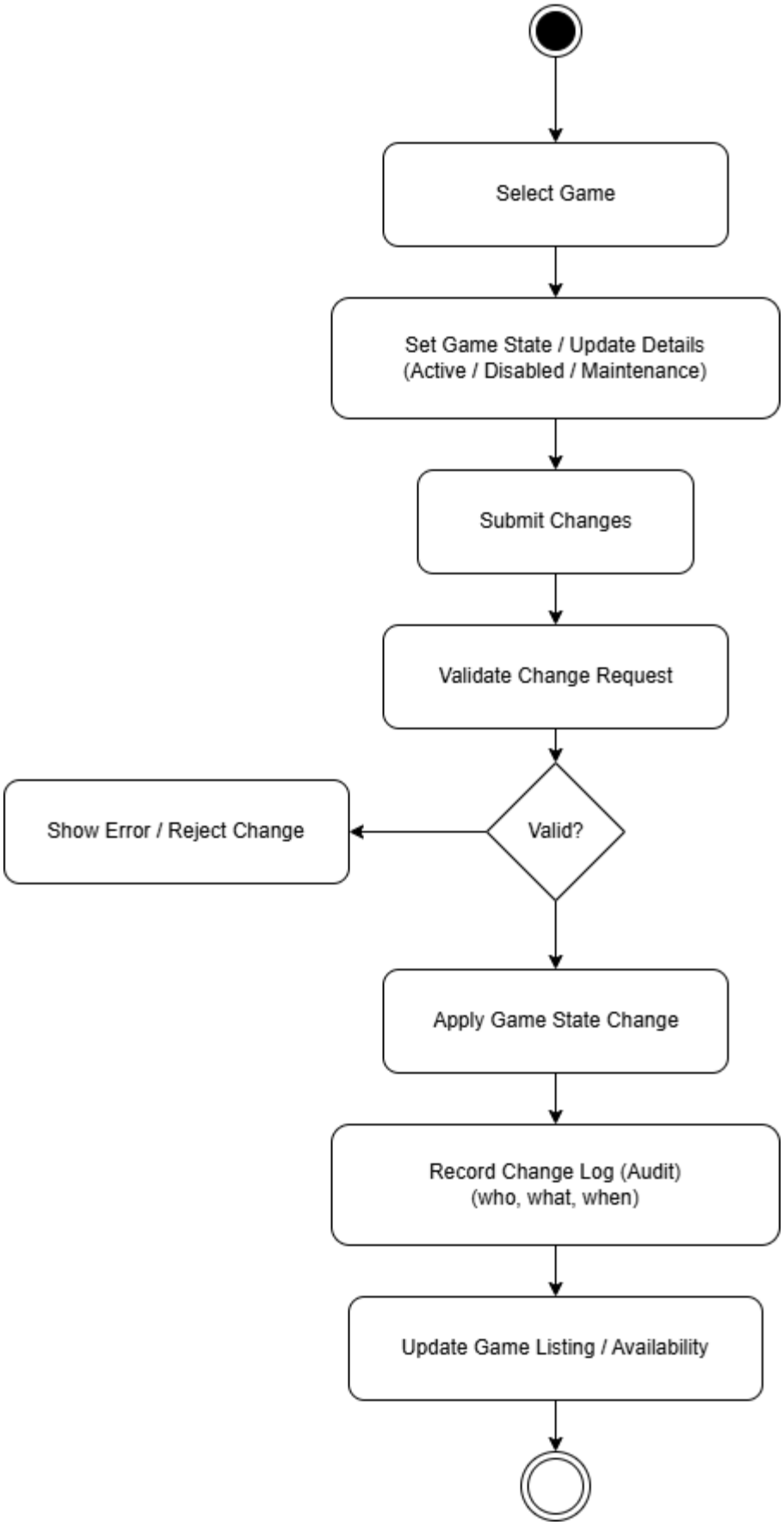
- Sequence Diagrams



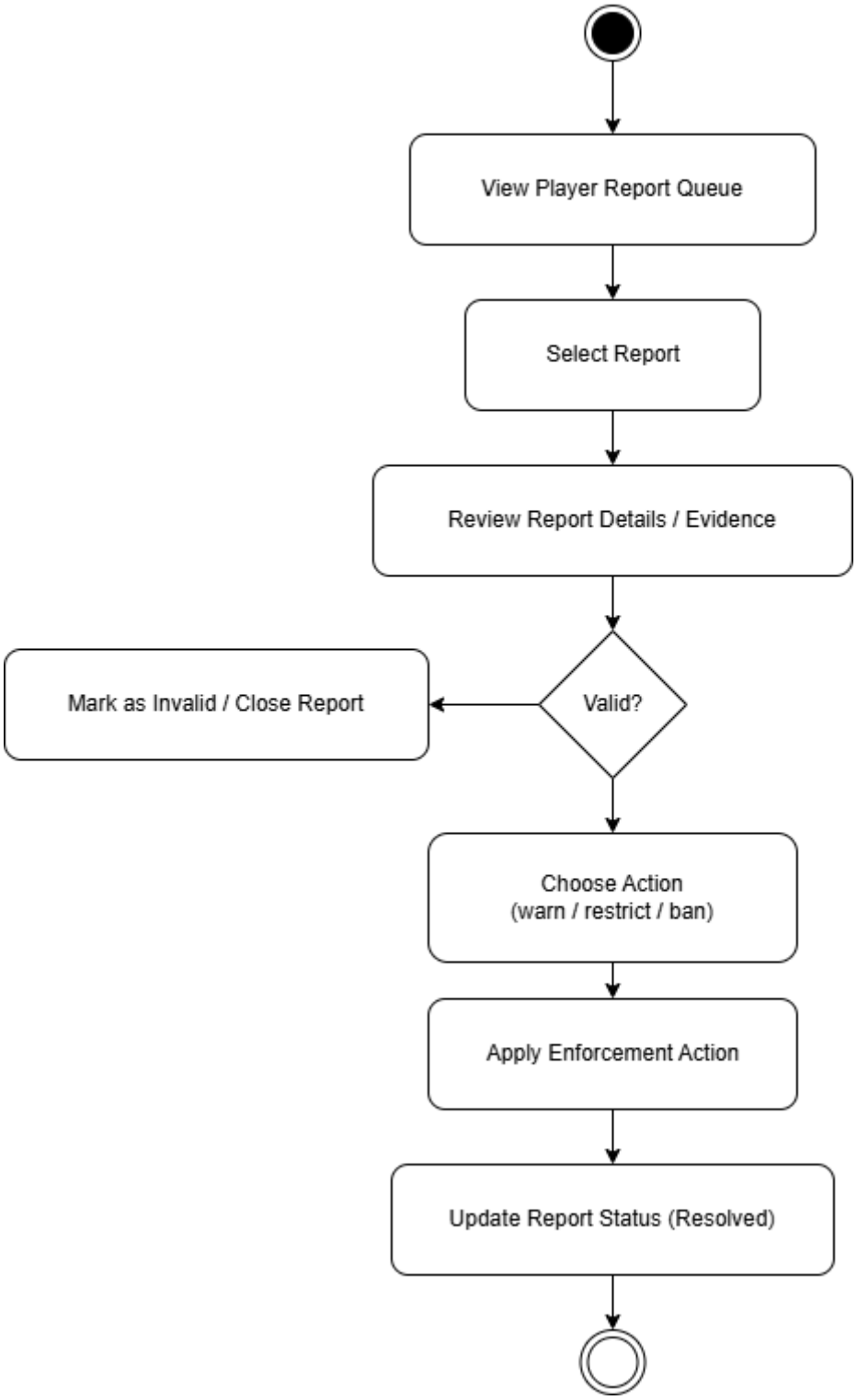

manage profile
Open Profile
Modify Profile Details
Submit Changes
Validate Details
Valid?
Show Error Message
Update Profile


publish game
Open Platform
Access Dashboard
Upload Game Details
Review required?
Yes
Submit Game for review
Modify Game
No
Approved?
yes
Publish the Game

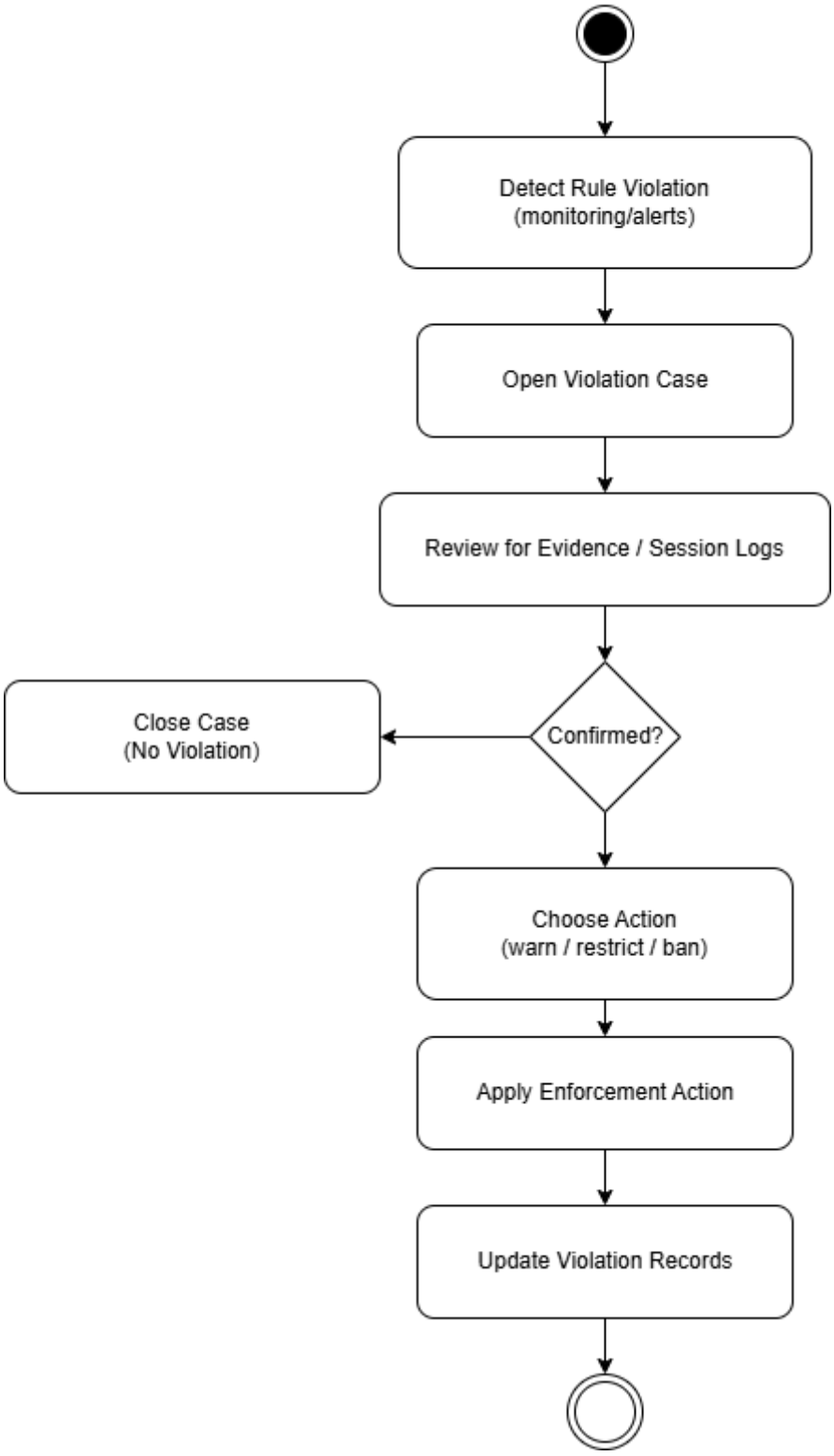
set game state



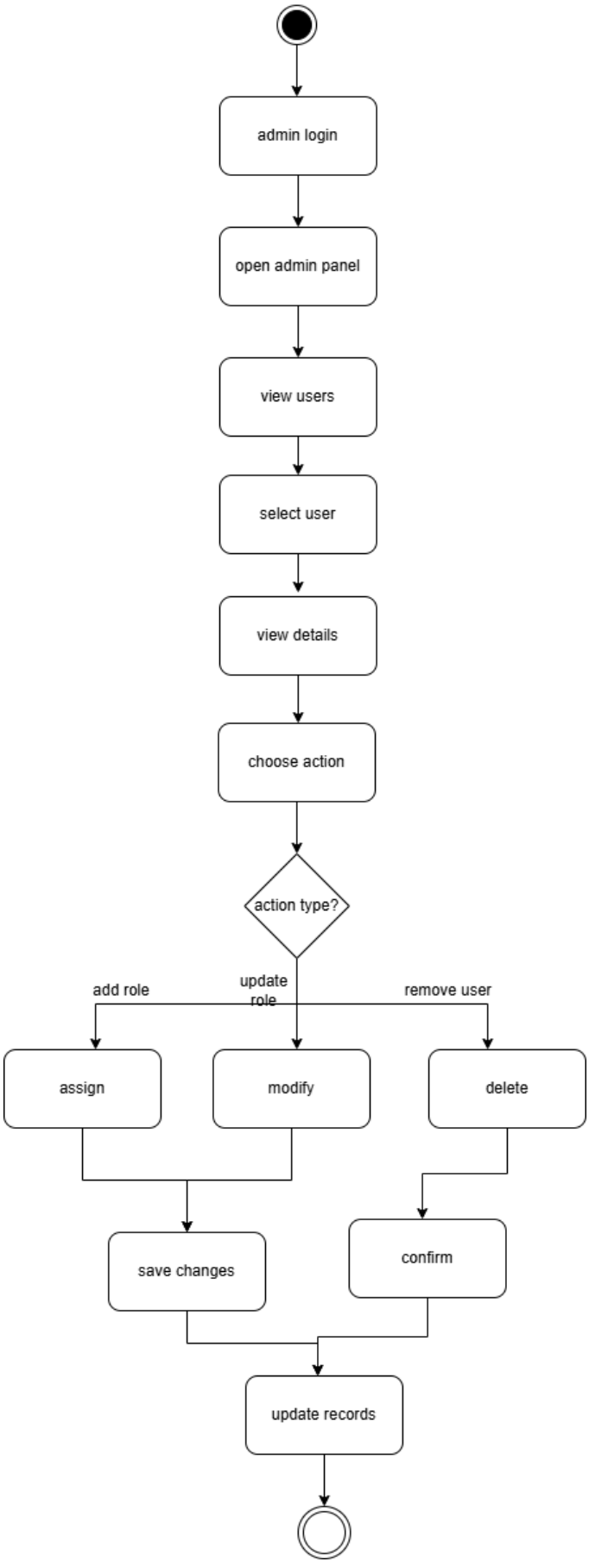
review player reports



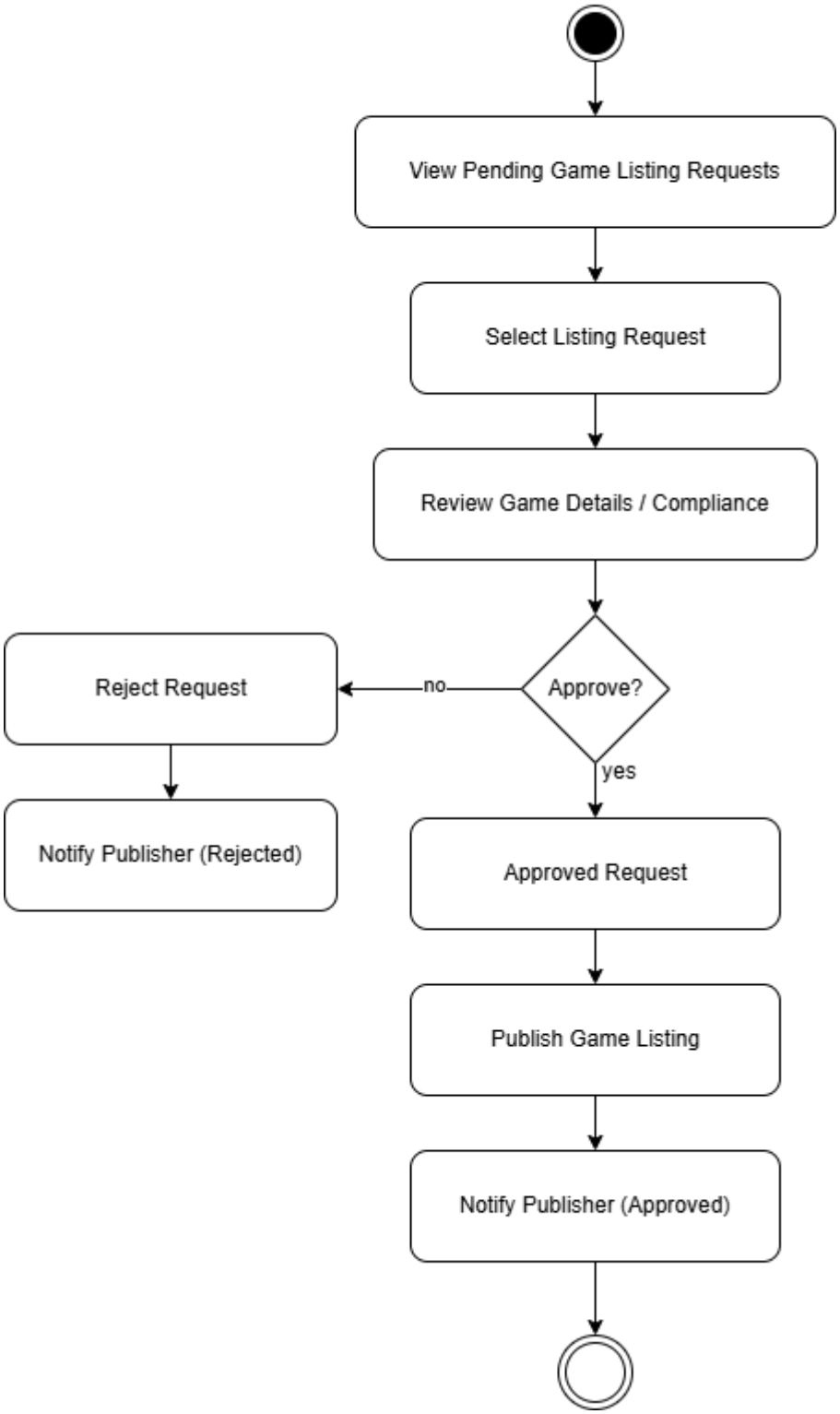
handle rule violations



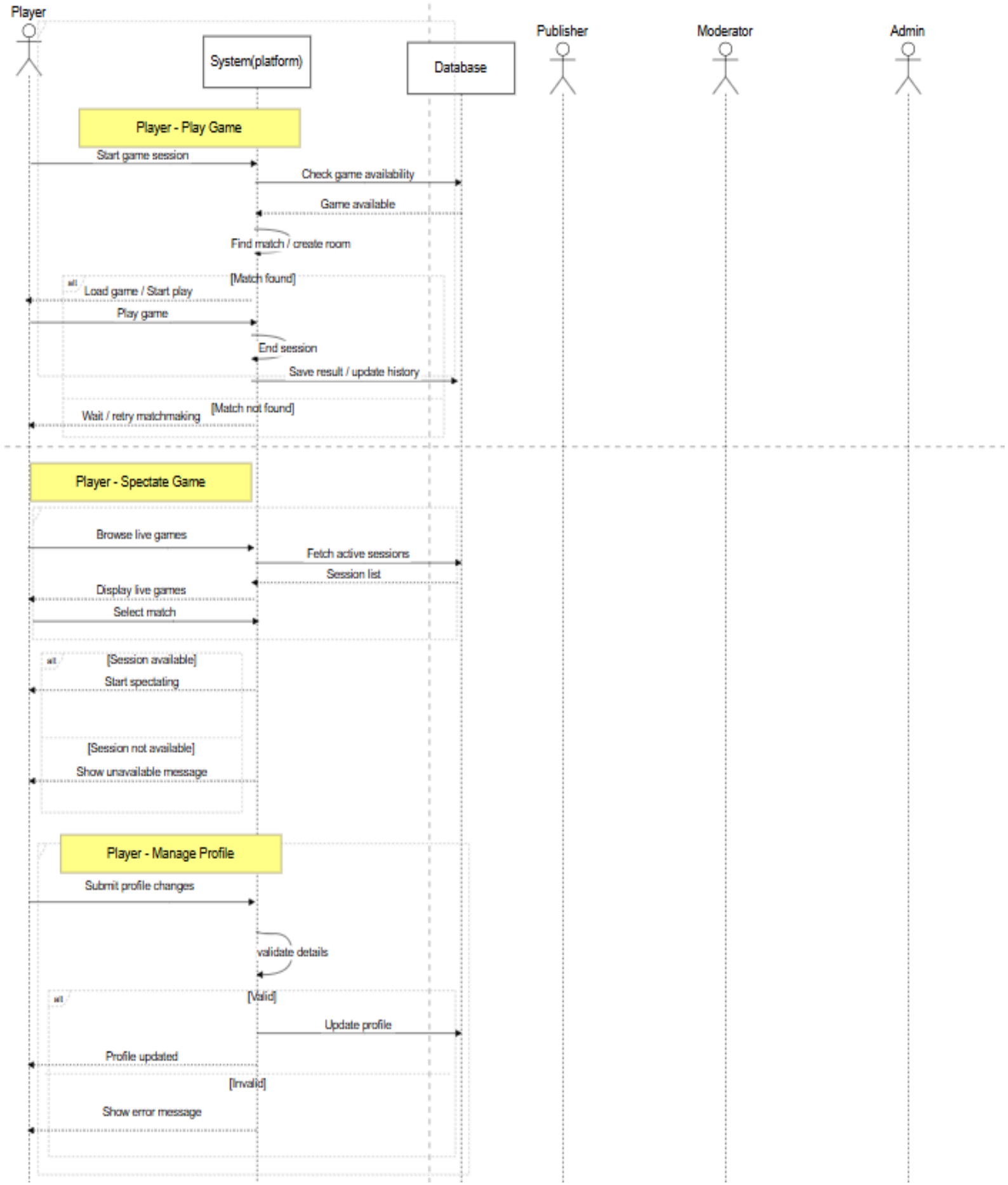
manage users and roles

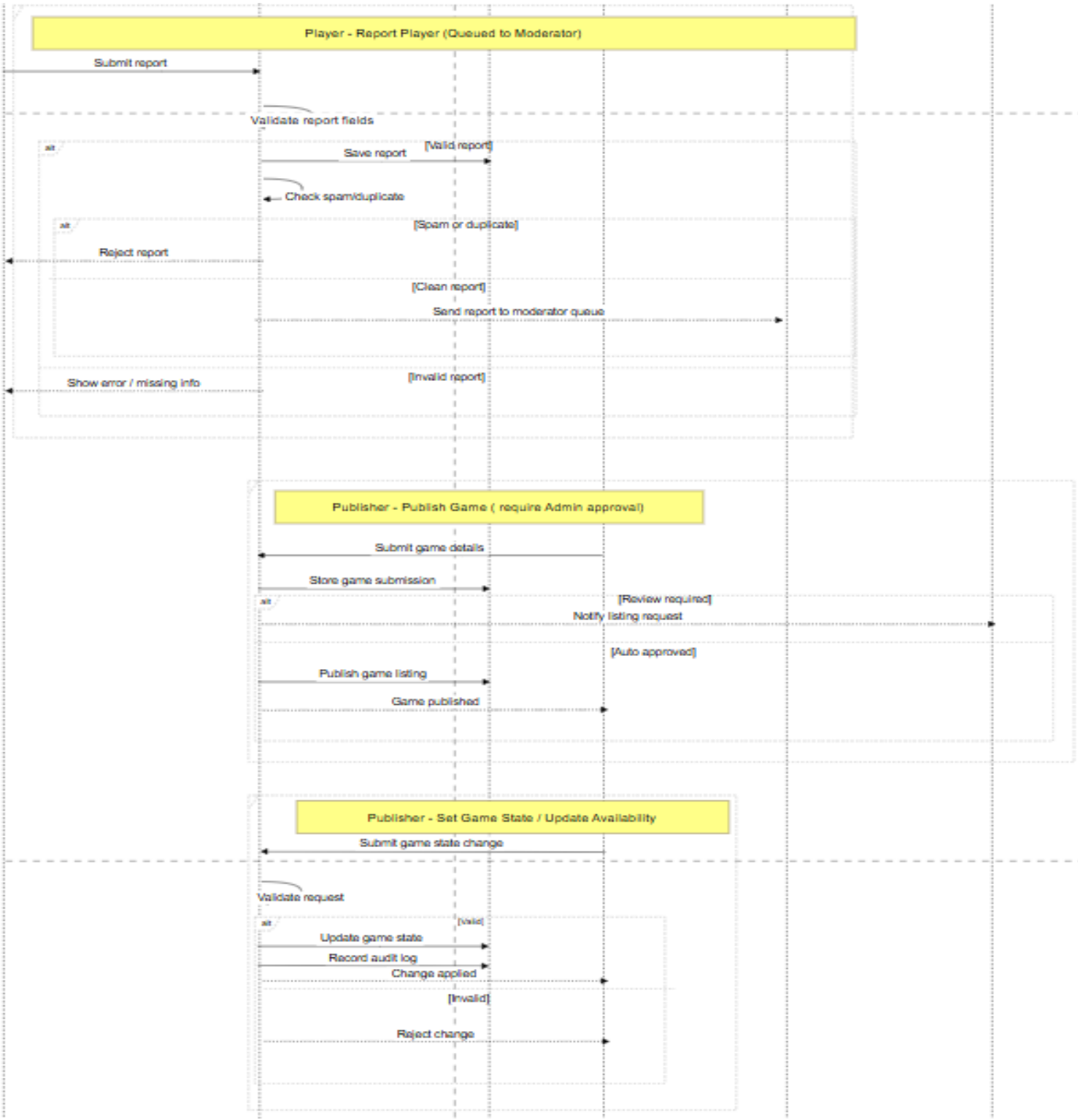


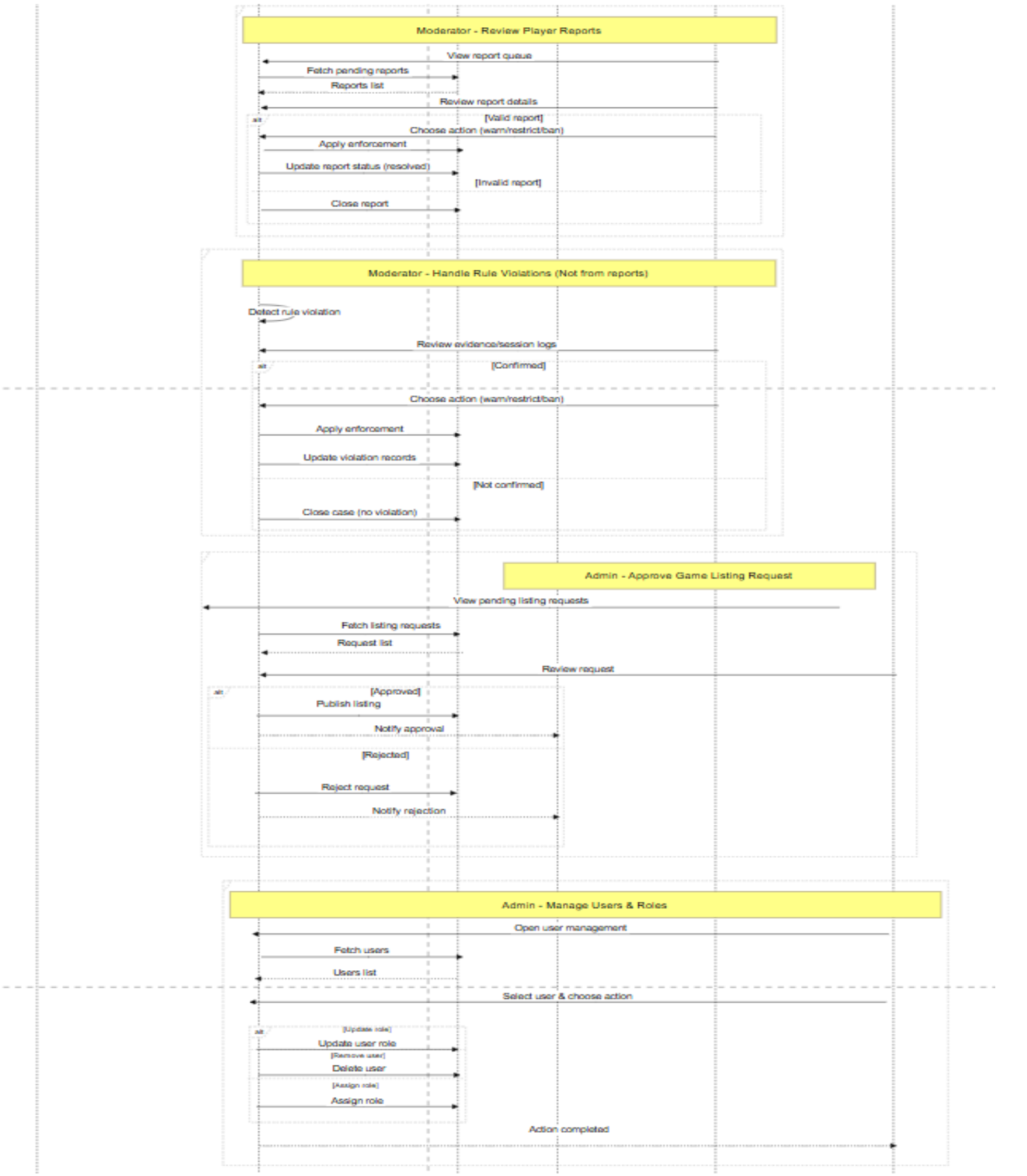
approve game listing request



Sequence diagram


Player
System(platform)
Database
Publisher
Moderator
Admin
Player - Play Game
Start game session
Check game availability
Game available
Find match / create room
[Match found]
Load game / Start play
Play game
End session
Save result / update history
[Match not found]
Wait / retry matchmaking
Player - Spectate Game
Browse live games
Fetch active sessions
Session list
Display live games
Select match
[Session available]
Start spectating
[Session not available]
Show unavailable message
Player - Manage Profile
Submit profile changes
validate details
[Valid]
Update profile
Profile updated
[Invalid]
Show error message


Player - Report Player (Queued to Moderator)
Submit report
Validate report fields
alt
[Valid report]
Save report
Check spam/duplicate
[Spam or duplicate]
Reject report
[Clean report]
Send report to moderator queue
[Invalid report]
Show error / missing info
Publisher - Publish Game (require Admin approval)
Submit game details
Store game submission
[Review required]
Notify listing request
[Auto approved]
Publish game listing
Game published
Publisher - Set Game State / Update Availability
Submit game state change
Validate request
[Valid]
Update game state
Record audit log
Change applied
[Invalid]
Reject change


Moderator - Review Player Reports
View report queue
Fetch pending reports
Reports list
Review report details
alt
[Valid report]
Choose action (warn/restrict/ban)
Apply enforcement
Update report status (resolved)
[Invalid report]
Close report
Moderator - Handle Rule Violations (Not from reports)
Detect rule violation
Review evidence/session logs
alt
[Confirmed]
Choose action (warn/restrict/ban)
Apply enforcement
Update violation records
[Not confirmed]
Close case (no violation)
Admin - Approve Game Listing Request
View pending listing requests
Fetch listing requests
Request list
Review request
alt
[Approved]
Publish listing
Notify approval
[Rejected]
Reject request
Notify rejection
Admin - Manage Users & Roles
Open user management
Fetch users
Users list
Select user & choose action
alt
[Update role]
Update user role
[Remove user]
Delete user
[Assign role]
Assign role
Action completed

These models describe interactions between actors and the system in a structured manner.

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8. System Evolution

8.1 Fundamental Assumptions

- The system is accessed via standard web interfaces.
- The number of users and games may increase over time.
- Platform rules and policies may evolve.

8.2 Anticipated Future Changes

- Addition of new game types and competitive modes.
- Enhanced analytics and monitoring features.
- Integration with external gaming services.

The system design should remain flexible to accommodate future expansion.

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9. Appendices

9.1 Hardware Requirements

Minimum: Basic server and internet connectivity
Optimal: Scalable cloud-based infrastructure

9.2 Database Requirements

- Centralized database for users, games, sessions, and reports
- Secure storage and data consistency

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10. Index

- Actor Index
- Function Index
- Diagram Index